

∴ The matrix is in echelon form but is not in reduced row echelon form.

Row reduce the matrices in Exercises 3 and 4 to reduced echelon form. Circle the pivot positions in the final matrix and in the original matrix and list the pivot columns.

3.
$$\begin{bmatrix} \textcircled{1} & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

Step 1: $\text{Row } 2 = \text{Row } 2 - 4 \times \text{Row } 1$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

Step 2: $\text{Row } 3 = \text{Row } 3 - 6 \times \text{Row } 1$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{bmatrix}$$

Step 3: $\text{Row } 2 = -\frac{1}{3} \times \text{Row } 2$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -5 & -10 & -15 \end{bmatrix}$$

Step 4: $\text{Row } 3 = \text{Row } 3 + 5 \times \text{Row } 2$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 5: $\text{Row } 1 = \text{Row } 1 - 2 \times \text{Row } 2$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

1. The zero row is below all non-zero rows. The condition that every zero row must be below all non-zero rows is True.

2. Leading entry of row 1 = 1 in Column 1
Leading entry of row 2 = 1 in Column 2

∴ The condition that the leading entry of a row must be in a column to the right of the columns of leading entries of all rows above it is True.